ST. PETERS HIGH SCHOOL -BUSABALA

COMPETENCE BASED ASSESSMENT 2026

BEGINNING OF TERM 1 EXAM 2026

S.6 BIOLOGY 1

TIME: 2 ½ HOURS

NAME…….. SIGN…….

INSTRUCTIONS

- **T**his paper consists of two Sections **A** and **B**.
- Respond to all items in Section **A** and to only Two items from Section **B.**
- Responses to Section A must be written in the **spaces** of the examination paper provided.

ADVICE

- Read each item carefully before you start your response.
- High quality of biology Language attracts higher scores.

ITEM 1

A research team in Busabala investigated energy production in two cell cultures: Culture X (muscle cells) and Culture Y (yeast cells). The cultures were grown in identical conditions except for oxygen availability.

After 2 hours, the following data were obtained:

Parameter	Culture X	Culture Y
Oxygen concentration	High	Low
Glucose consumed (mg)	18	40
ATP produced (units)	72	24
End-products detected	$CO_2 + H_2O$	Ethanol + CO_2

Task

a) Explain why Culture Y consumed more glucose than Culture X.

(b) What does this suggest about the efficiency of the metabolic pathway used by Culture Y?

(c) Using the data, compare ATP yield per mg of glucose in cultures X and Y.

(d)i Explain the differences in end-products observed in the two cultures.

(ii) Describe one structural feature of mitochondria that enhances efficiency of the process occurring in Culture X.

Item 2

A farmer in Busabala planted two varieties of beans, Variety A and Variety B, under the same environmental conditions. After 6 weeks, leaf samples were analysed for stomatal density and chlorophyll content.

Parameter	Variety A	Variety B
Habitat of origin	Arid region	Wet region
Stomatal density (per mm^2)	120	380
Guard cell thickness	Thick	Thin
Chlorophyll content (mg/g)	3.2	5.8
Leaf surface	Waxy	Smooth

Task

(a) Using the data, explain the adaptations of Variety A to survive in its native habitat.

(b) Suggest why Variety B has a higher chlorophyll content than Variety A.

(c) Predict how each variety would respond to prolonged drought and justify your answer.

(d) Explain how stomatal density influences the rate of transpiration in the two varieties.

SECTION B

(Choose one item from each part)

PART 1

ITEM 3

A hospital monitored two patients, Patient P and Patient Q, during a glucose tolerance test. Blood glucose levels (mmol/L) were recorded over time.

Time after glucose intake	0 min	30 min	60 min	120 min	180 min
Patient P	5.2	7.8	6.0	5.4	5.3
Patient Q	5.4	11.0	13.5	12.8	11.5

Task

(a) Describe the normal homeostatic response in Patient P after glucose intake.
(b) Using data, explain the abnormal pattern in Patient Q.
(c) What hormone is likely deficient or malfunctioning in Patient Q? Explain its normal role.

(d) Propose one long-term complication of Patient Q's condition and explain its physiological basis

Item 4

A study investigated thermoregulation in goats from **Highland Region (H goats)** and **Lowland Region (L goats)** when exposed to increasing environmental temperatures

Temperature (°C)	**20**	**30**	**35**	**40**
Mean respiration rate (breaths/min) – H goats	18	24	29	34
Mean respiration rate (breaths/min) – L goats	22	33	40	52

Task

(a) Interpret the data and compare thermal tolerance of the two goat populations.
(b) Explain why L goats show a sharper increase in respiration rate at higher temperatures.
(c) Describe two physiological mechanisms goats use to lose heat.
(d) Suggest how natural selection may have shaped the differences observed between the two populations

PART 11 (Choose one item)

Item 5

A population of beetles was studied for colour variation. The environment changed when dark volcanic ash covered the ground. Population changes were recorded:

Generation	**Light beetles (%)**	**Dark beetles (%)**
1	70	30
5	40	60
10	15	85

Task

(a) Describe the evolutionary process responsible for the change in beetle color frequencies.
(b) Explain how the environmental change exerted selective pressure.
(c) Predict what would happen if the environment reverted to light-colored soil.
(d) State one assumption required for natural selection to result in such changes

Item 6

A maize breeder studied two genes:

- **Gene A**: kernel texture (dominant = smooth, recessive = wrinkled)
- **Gene B**: kernel colour (dominant = purple, recessive = yellow)

A dihybrid cross was performed between two heterozygous plants: **AaBb × AaBb**. The observed phenotypic ratio among 800 seeds was:

Phenotype	Observed number
Smooth purple	360
Smooth yellow	120
Wrinkled purple	100
Wrinkled yellow	220

Task

(a) Calculate the expected Mendelian ratio and expected numbers for each phenotype.
(b) Using chi-square analysis principles, explain whether the observed results deviate significantly from expectation.
(c) Suggest a biological reason that could explain the deviation from Mendelian ratios.
(d) Explain why dihybrid crosses illustrate independent assortment.

END

- **"You are braver than you believe, stronger than you seem and smarter than you think." ...**